

CHAPAIS, QC, Canada

WMO: 718240

Lat: 49.8225N

Lon: 74.9753W

Elev: 381

StdP: 96.83

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-32.1	-39.5	0.1	-35.0	-36.1	0.1	-32.0	5.5	-12.0	4.9	-13.6	1.1	210	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	28.1	17.8	26.1	16.7	24.2	16.1	20.1	24.3	19.1	22.9	18.0	21.8	2.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.3	21.3	17.7	13.3	20.6	16.6	12.4	19.7	59.6	24.6	55.9	22.8	52.3	21.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.3	3.9	-39.9	31.5	2.4	1.5	-41.7	32.6	-43.1	33.5	-44.5	34.4	-46.2	35.5
			-40.2	22.4	2.6	1.3	-42.1	23.3	-43.7	24.1	-45.1	24.8	-47.0	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-18.0	-16.3	-9.7	-0.9	8.2	14.0	16.4	15.1	10.8	3.8	-4.1	-13.0	(6)
(7)		DBStd	13.47	8.28	7.59	7.80	5.87	5.42	4.37	3.40	3.49	4.51	4.79	5.63	7.37	(7)
(8)		HDD10.0	4038	867	737	609	327	102	12	1	3	43	203	423	712	(8)
(9)		HDD18.3	6509	1126	970	868	576	317	141	79	110	228	452	673	971	(9)
(10)		CDD10.0	621	0	0	1	1	46	133	200	163	68	9	0	0	(10)
(11)		CDD18.3	49	0	0	0	0	2	12	21	11	3	0	0	0	(11)
(12)		CDH23.3	585	0	0	0	0	55	185	226	88	31	0	0	0	(12)
(13)		CDH26.7	126	0	0	0	0	13	52	48	10	3	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.8	1.8	2.0	2.0	1.9	1.8	1.7	1.6	1.7	1.8	1.9	1.7	(14)
(15)	Precipitation	PrecAvg	987	55	43	48	57	73	97	116	107	122	97	91	68	(15)
(16)		PrecMax	1213	99	88	98	94	137	165	187	232	284	171	170	135	(16)
(17)		PrecMin	806	13	0	0	12	19	34	42	41	61	37	35	1	(17)
(18)		PrecStd	114	23	20	24	21	30	38	41	44	49	35	31	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	2.2	11.1	17.3	28.2	31.0	30.4	28.1	26.5	20.3	10.5	3.6	(19)
(20)			MCWB	2.5	0.8	6.1	8.9	16.7	18.4	18.9	19.6	17.7	14.1	8.6	3.1	(20)
(21)		2%	DB	0.5	0.0	6.9	13.8	24.4	28.0	27.9	25.7	23.6	16.7	7.6	0.8	(21)
(22)			MCWB	0.3	-0.5	2.6	6.4	13.2	16.8	18.1	17.8	17.1	12.3	6.2	0.6	(22)
(23)		5%	DB	-2.5	-2.1	3.9	10.5	21.3	25.2	25.9	23.8	21.0	13.7	5.0	-0.5	(23)
(24)			MCWB	-3.0	-2.8	0.7	4.8	12.1	15.7	16.9	16.8	15.9	11.1	4.2	-0.8	(24)
(25)		10%	DB	-5.3	-4.2	1.6	7.6	18.3	22.8	23.8	21.9	18.7	11.1	2.9	-2.4	(25)
(26)			MCWB	-5.7	-5.0	-0.4	3.3	10.6	14.6	16.5	16.3	14.8	9.2	2.2	-2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	0.9	7.7	10.6	18.6	21.0	21.7	21.1	19.8	15.4	9.5	3.4	(27)
(28)			MCDWB	2.8	1.9	8.8	14.4	23.3	26.0	25.6	25.1	24.1	19.2	10.2	3.6	(28)
(29)		2%	WB	0.4	-0.8	3.4	7.5	15.5	19.1	20.1	19.7	18.4	13.1	6.7	0.7	(29)
(30)			MCDWB	0.6	-0.2	5.9	11.5	20.6	24.1	24.3	23.1	21.6	15.3	7.3	0.9	(30)
(31)		5%	WB	-3.1	-2.9	1.5	5.7	13.4	17.6	18.9	18.7	16.9	11.1	4.2	-0.7	(31)
(32)			MCDWB	-2.7	-2.2	3.6	9.7	18.8	22.4	23.1	21.9	20.1	12.8	5.0	-0.5	(32)
(33)		10%	WB	-6.0	-5.1	-0.2	4.0	11.8	16.3	17.8	17.4	15.3	9.1	2.2	-2.7	(33)
(34)			MCDWB	-5.5	-4.5	1.7	7.6	17.2	21.1	21.7	20.5	18.4	10.5	2.9	-2.4	(34)
(35)	Mean Daily Temperature Range	MDBR		11.0	12.1	13.3	11.4	13.0	13.2	12.1	10.9	10.7	7.4	6.6	9.0	(35)
(36)		5% DB	MCDBR	10.0	9.2	14.4	15.4	17.6	17.3	15.7	14.1	13.8	12.1	8.2	7.3	(36)
(37)			MCWBR	9.9	8.4	10.7	9.1	8.6	7.9	7.8	7.5	8.4	8.1	7.3	7.5	(37)
(38)			MCDBR	10.2	8.7	12.1	13.4	15.0	13.3	12.0	11.1	12.0	9.9	8.0	7.8	(38)
(39)		5% WB	MCWBR	10.2	8.2	9.9	9.0	8.8	7.2	7.4	7.3	8.0	7.4	7.4	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.247	0.263	0.279	0.329	0.337	0.356	0.373	0.354	0.348	0.304	0.280	0.243	(40)
(41)		taud		2.285	2.257	2.263	2.339	2.476	2.452	2.424	2.481	2.466	2.581	2.487	2.340	(41)
(42)		Ebn at Noon		814	888	934	917	922	902	879	883	850	842	778	770	(42)
(43)		Edh at Noon		68	90	108	114	104	107	109	98	90	67	57	56	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.22	2.29	3.80	4.82	5.24	5.62	5.11	4.18	3.17	1.78	0.89	0.73	(44)
(45)		RadStd		0.15	0.29	0.35	0.38	0.49	0.43	0.49	0.29	0.30	0.21	0.11	0.11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page